ENVARIABELSANALYS DEL 2
SEMINARIUM 23: DIFF-EKVATIONER

Första ordningens diff-ek

Ex: Lös begynnelsevärdesproblem

$$\begin{cases} (1-x^2)y'(x) + xy(x) = (1-x^2)^{3/2} \arcsin(x), & |x|<1 \\ y(0) = 2 \end{cases}$$

OBS: $|x| \iff \pm 1$

$|x|<1 \iff |x-0|<1 \iff -1<x<1$

- om $f(x) = \arcsin(x)$ blir $f'(x) = \frac{1}{\sqrt{1-x^2}}$
- $(1-x^2)y' + xy = (1-x^2)^{3/2}\arcsin(x)$

$/ : (1-x^2)$

GER: $y' + \frac{xy}{1-x^2} = \frac{(1-x^2)^{3/2}}{1-x^2} \arcsin(x)$

$y' + \frac{xy}{1-x^2} = \frac{1\cancel{(1-x^2)}\sqrt{1-x^2}}{1\cancel{(1-x^2)}} \arcsin(x)$

$y' + \frac{xy}{1-x^2} = \sqrt{1-x^2}\arcsin(x)$

SÖK INTEGRERANDE FAKTOR

$e^{\int \frac{x}{1-x^2}dx}$ | $\int \frac{x}{1-x^2}dx = \int -\frac{1}{2} \cdot \frac{-2x}{1-x^2}dx =$

$= -\frac{1}{2}\int \frac{-2x}{1-x^2}dx = -\frac{1}{2}\ln|1-x^2|$ $\left[\begin{matrix} |x|<1 \iff \\ 1-x^2>0 \end{matrix} \right]$

$= -\frac{1}{2}\ln(1-x^2)$

TILLBAKA:

$e^{-\frac{1}{2}\ln(1-x^2)} = e^{\ln(1-x^2)^{-1/2}} = \left[e^{\ln a} = a\right]$

$= (1-x^2)^{-1/2} = \frac{1}{\sqrt{1-x^2}}$

MULTIPLICERA IN I.F.

$y' \frac{1}{\sqrt{1-x^2}} + \frac{x}{1-x^2} \cdot \frac{1}{\sqrt{1-x^2}} \cdot y = \cancel{\sqrt{1-x^2}} \cdot \frac{1}{\cancel{\sqrt{1-x^2}}} \arcsin(x)$,

SÅ: $\left(y' \frac{1}{\sqrt{1-x^2}}\right)' = \arcsin(x)$ $\longrightarrow$

INTEGRERA BÅDA LEDEN:

$$y \cdot \frac{1}{\sqrt{1-x^2}} = \int \arcsin(x)\,dx$$

$$\cdot \int \arcsin(x)\,dx = \begin{bmatrix} u = \arcsin(x) & u' = \frac{1}{\sqrt{1-x^2}} \\ & v' = 1 \\ v = x & \end{bmatrix}$$

$$= \arcsin(x) \cdot x - \int \frac{x}{\sqrt{1-x^2}}\,dx = \quad t = 1-x^2$$

$$dt = -2x\,dx$$

$$-\frac{1}{2}\,dt = x\,dx$$

$$= x\arcsin(x) + \frac{1}{2}\int \frac{1}{\sqrt{t}}\,dt$$

$$= x\arcsin(x) + \frac{1}{2} \cdot \int t^{-1/2}\,dt$$

$$= x\arcsin(x) + \frac{1}{2}\,\frac{t^{1/2}}{1/2} + C = x\arcsin(x) + \sqrt{t} + C$$

Om $t = 1-x^2$ $\Rightarrow$ $x\arcsin(x) + \sqrt{1-x^2} + C$

TILLBAKA

$$y\,\frac{1}{\sqrt{1-x^2}} = x\arcsin(x) + \sqrt{1-x^2} + C$$

$$y = \sqrt{1-x^2} \cdot x\arcsin(x) + \sqrt{1-x^2}\sqrt{1-x^2} + C\sqrt{1-x^2}$$

$$y = \sqrt{1-x^2}\,x\arcsin(x) + (1-x^2) + C\sqrt{1-x^2}$$

DETTA ÄR DÅ EN ALLMÄN LÖSNING

ANVÄND BIVILLKORET

VI SÖKER DÅ $y(0) = 2$

för $x = 0$ och $y = 2$ ges =

$$2 = 0\sqrt{1-0^2} \cdot \underbrace{\arcsin(0)}_{=0} + 1 - 0^2 + C\sqrt{1-0^2}$$

$$2 = 0 + 1 - 0 + C \quad \Leftrightarrow$$

$C = 1$ | FÖR $C=1$ BLIR LÖSNINGEN:

$$y = \sqrt{1-x^2}\,x\arcsin(x) + (1-x^2) + \sqrt{1-x^2}$$

SVAR:

EX: Bestäm den LÖSNING TILL diff.ek

$$(1+x^2)y' - 2xy = (1+x^2)^2 \arctan(x)$$

som uppfyller $y(0) = 1$

LÖSNING:

$$(1+x^2)y' - 2xy = (1+x^2)^2 \arctan(x) \quad /:(1+x^2)$$

$$y' - \frac{2x}{1+x^2} \cdot y = \frac{\cancel{(1+x^2)^2}^{1} \arctan(x)}{\cancel{(1+x^2)}_{1}} =$$

$$= y' - \frac{2x}{1+x^2} \cdot y = (1+x^2)\arctan(x)$$

SÖK INTEGRERANDE FAKTOR

$$e^{\int \frac{2x}{1+x^2}dx} = \left[\int \frac{2x}{1+x^2}dx = \ln(1+x^2) \right]$$

$$= e^{\ln(1+x^2)}$$

$$= \underline{\underline{(1+x^2)}}$$

MULTIPLICERA MED I.F.

$$y'(1+x^2) - 2 \cdot \frac{x\cancel{(1+x^2)}}{\cancel{(1+x^2)}} y = (1+x^2)^2 \arctan(x)$$

$$= y'(1+x^2) - 2xy = (1+x^2)\arctan(x)$$

$$\left(y(1+x^2)\right)' = (1+x^2)^2 \arctan(x)$$

$$y(1+x^2) = \int (1+x^2)^2 \arctan(x)\,dx$$

$$y(1+x^2 = \int\left((1^2 + 2x^2 + x^4)\arctan(x)\right)dx$$

$$y(1+x^2) = \int \arctan(x)dx + 2x^2\arctan(x) + x^4\arctan(x)dx$$

$$\left[\begin{array}{ll} u = \arctan(x) & u' = \frac{1}{1+x^2} \\ v = x + \frac{2x^3}{3} + \frac{x^5}{5} & v' = 1 + 2x^2 + x^4 \end{array} \right]$$

$$\arctan(x)\left(x + 2\frac{x^3}{3} + \frac{x^5}{5}\right) - \int\left(\frac{1}{1+x^2}\right)\left(x + \frac{2x^3}{3} + \frac{x^5}{5}\right)dx$$

MAGGANS Lösning

$$y' - \frac{2x}{1+x^2}\, y = (1+x^2)\arctan(x)$$

I.F
$$e^{\int -\left(\frac{2x}{1+x^2}\right)dx} = e^{-\int \frac{2x}{1+x^2}dx} = e^{-\ln(1+x^2)}$$

$$= e^{\ln(1+x^2)^{-1}} = \boxed{\frac{1}{1+x^2}}$$

Multiplicera:

$$y' \cdot \frac{1}{1+x^2} - \frac{1}{1+x^2}\left(\frac{-2}{1+x^2}\right) y = \frac{\cancel{(1+x^2)}}{\cancel{(1+x^2)}}\arctan(x)$$

$$= y' \cdot \frac{1}{1+x^2} + \frac{1}{(1+x^2)^2}\, y = \arctan(x)$$

$$\left(y \cdot \frac{1}{1+x^2}\right)' = \arctan(x)$$

$$y\,\frac{1}{1+x^2} = \int \arctan(x)$$

$$\cdot \int \arctan(x) \cdot 1 = \begin{bmatrix} u = \arctan(x) & u' = \frac{1}{1+x^2} \\ v = x & v' = 1 \end{bmatrix}$$

$$= x\arctan(x) - \int \frac{x}{1+x^2}dx = x\arctan(x) - \frac{1}{2}\int \frac{2x}{1+x^2}dx =$$

$$= x\arctan(x) - \frac{1}{2}\ln(1+x^2) + C$$

Så: $y \cdot \frac{1}{1+x^2} = x\arctan(x) - \frac{1}{2}\ln(1+x^2) + C$

$$y = (1+x^2)\,x\arctan(x) - \frac{1}{2}\ln(1+x^2)(1+x^2) + C(1+x^2)$$

$$P(0,1) \Rightarrow 1 = \arctan(0)\cdot 0 - \frac{1}{2}\underbrace{\ln(1+0)}_{0}(1+0) + C(1+0)$$

$$= 1 = C \quad \Leftrightarrow$$

SVAR:

$$y = (1+x^2)\,x\arctan(x) - \frac{1}{2}\ln(1+x^2)(1+x^2) + C(1+x^2)$$